

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

OCT 2022 TRIMESTER

FINAL ASSESSMENT

MPU32143 ENGLISH FOR INFORMATION TECHNOLOGY

TIME: XX (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) COMPUTER
ENGINEERING (CT)

BACHELOR OF COMPUTER SCIENCE (HONS) (CS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING (CN)

BACHELOR OF INFORMATION SYSTEMS (HONOURS) INFORMATION
SYSTEMS ENGINEERING (IA)

BACHELOR OF INFORMATION SYSTEMS (HONS) BUSINESS
INFORMATION SYSTEMS (IB)

BACHELOR OF INFORMATION SYSTEMS (HONOURS) DIGITAL ECONOMY
TECHNOLOGY (DE)

Instructions to Candidates:

There are **THREE (3)** sections in this paper – Section A, B and C

SECTION A [TOTAL: 40 MARKS]

Reading Comprehension

Answer **ALL** questions.

SECTION B [TOTAL: 30 MARKS]

Report Writing

Answer **ONE (1)** question **ONLY**.

SECTION C [TOTAL: 30 MARKS]

Interpretation of Graphical Data

Answer **THE** question.

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SECTION A [TOTAL: 40 MARKS]

READING COMPREHENSION

Read carefully the text below and answer **ALL** the following questions.

The pandemic has changed how universities' view Cybersecurity by Maxime Bilodeau

Are Canadian universities vulnerable to cyberattacks? Not according to them. Unless 1
regular operations are affected, very few postsecondary institutions report being victims
of cybercrime. When a major security flaw was discovered in December 2021, it was not
until the Quebec government demanded that all exposed sites, services and systems shut
down as a “preventative measure” that many Quebec universities acknowledged it. The
University de Montréal’s response was similar when it was the subject of a global
ransomware attack a few years ago.

This relative radio silence gives the impression that public institutions like universities are 2
immune to cyber threats. But experts consulted by University Affairs say the reality is
likely quite different. *“The little data we have comes largely from British cyber security
centre reports, which show just how vulnerable universities are to these very real threats,”*
said Benoît Dupont, a professor in UdeM’s school of criminology and Canada Research
Chair in cyber security. Jisc, a U.K.-based organization whose mission is to digitally
empower the country’s higher education institutions, conducts a survey every year to see
where things stand. The most recent results, published in November 2021, found that
approximately 60 per cent of the 166 colleges and universities surveyed had reported a
cyberattack in the previous 12 months. The survey also found a huge increase in
ransomware attacks in the higher education sector.

The COVID-19 pandemic has contributed to this phenomenon. The public health crisis 3
forced Canadian universities to move everything online. Their IT infrastructure became a
lifeline almost overnight, making them more attractive targets. “The risk of cyberattack
was high before March 2020 and it is just as high now. The COVID curveball has made
the university community aware of its own vulnerability and the importance of developing
responsible digital behaviour,” explained Mohammad Mannan, Associate Professor at
Concordia University’s Institute for Information Systems Engineering.

On Feb. 17, the Government of Canada announced the creation of the Cyber Security 4
Innovation Network, which will be led by the National Cybersecurity Consortium. The
consortium was created in 2020 by the centres of expertise in cybersecurity at Concordia
University, Ryerson University, University of Calgary, University of New Brunswick
and University of Waterloo. This newly created network will receive \$80 million dollars
in funding over four years, and will have the mandate of “enhancing research and

SECTION A (CONTINUED)

development, increase commercialization, and developing skilled cybersecurity talent across Canada,” as explained in the press release announcing this initiative. Another challenge is that institutions have to balance security concerns with their mission. Practices rooted in academic freedom mean larger networks that are by nature more exposed. “This culture of openness and transparency is vastly different compared to private sector businesses,” said Nora Boulahia Cuppens, a professor in the department of computer engineering and software engineering at Polytechnique Montréal. She believes we have to stop thinking of cybercriminals as scammers hiding behind their keyboards. Cybercrime can also be an inside job.

Now more than ever, research relies on national and international partnerships between a multitude of stakeholders, including researchers, organizations and research centres. But would-be attackers could be hiding under the guise of collaboration. That is why the Canadian government took action in March 2021 by introducing National Security Guidelines for Research Partnerships to protect Canadian research and intellectual property against foreign interference, spying and theft. Going forward, applications for funding under the Alliance grant program administered by the Natural Sciences and Engineering Research Council of Canada (NSERC) for research partnerships involving one or more private-sector organization must also include a risk assessment. Minister of Innovation, Science and Industry François-Philippe Champagne announced in a press release that projects that are deemed high risk, or where the risk cannot be mitigated, will not be funded. 5

For now, the guidelines only apply to a small number of grants in research areas considered “strategic,” such as artificial intelligence, nanotechnology and biofabrication. But it is only a matter of time before they are expanded to include other fields, according to Dr. Dupont. Research teams will be held accountable for risk management, and cybersecurity in particular. Sooner or later, universities will have no choice but to get on board. And some may have a lot of catching up to do. “The methods universities use to take action on data protection and cybersecurity are limited. Awareness campaigns just aren’t going to cut it,” said Dr. Dupont. Universities can start by investing in simple safeguards, like setting up multifactor authentication for those that use their IT infrastructure, as well as routinely using virtual private networks to access sensitive data from outside the institution. 6

And in the long term? Everyone in academia must take responsibility for cybersecurity – starting with students, who make up a majority of the university community. “They are agents for change. They’re the ones who will be dealing with tomorrow’s threats, like quantum computing and 5G, which will increase the potential target areas,” warned Dr. Cuppens, who co-organized an ethical hacking competition in early February. “It’s a way to get our students thinking about cybersecurity issues,” she said. 7

Adapted from : <https://www.universityaffairs.ca/news/news-article/universities-need-to-start-taking-cybersecurity-seriously>

Answer **ALL** the following questions.

Q1. Identify **ONE (1) word** which carries the similar meaning based on the context.

- a) preventative (paragraph 1) (1 mark)
- b) empower (paragraph 2) (1 mark)
- c) aware (paragraph 3) (1 mark)
- d) vulnerability (paragraph 3) (1 mark)
- e) vastly (paragraph 3) (1 mark)
- f) mandate (paragraph 4) (1 mark)
- g) administered (paragraph 5) (1 mark)
- h) accountable (paragraph 6) (1 mark)
- i) authentication (paragraph 6) (1 mark)
- j) threats (paragraph 7) (1 mark)

[Total: 10 marks]

Q2. What are the methods universities can use to take action on data protection and cybersecurity?
(6 marks)

Q3. “*Cybercrime can also be an inside job*” (Paragraph 4)

In your own words, describe the intended meaning of the phrase stated above.

(2 marks)

Q4. How did the Canadian government prevent the appearance of *would-be attackers*?

(2 marks)

Q5. In your own words, provide **FIVE (5) advantages and FIVE (5) disadvantages** of cybersecurity. Provide **ONE (1) elaboration** for each advantage and disadvantage.

(20 marks)

[TOTAL: 40 MARKS]

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SECTION B [TOTAL: 30 MARKS]

REPORT WRITING

Answer **ONE (1)** question **ONLY**

Q1. You are the president of the Young Entrepreneurs Society at your university. The society will be organising a food festival “**Feast- A- Licious**” for 3 days. The purpose of the festival is to raise funds and the entire amount will be used for charitable causes. The festival will take place from 20th December 2022 to 23rd December 2022.

Write a report explaining the progress of the preparations for the festival to the advisory lecturer of the society, Dr. Lydia Hunt to give some updates on the preparation of the “**Feast- A- Licious**” festival. Include details on the tasks completed and also the remaining tasks as well as any hindrances you have faced so far.

Your report should include the following points:

Completed Work

Venue

On-going work

Food vendors/hawkers

Future Work

Special gifts for volunteers and vendors

Add another **THREE (3) relevant points** about the event preparation and provide sufficient elaboration. The body of the report **should not be more than 250 words.**

(30 marks)

[TOTAL: 30 MARKS]

Q2. You are the Assistant Manager at IT Technical Specialist Sdn. Bhd. Recently you received many complaints from the HR department and other departments on the issue of Internet abuse in the workplace. Based on the investigation conducted last month, there are top three complaints which result in decreased employees' productivity during working hours.

- Illegal downloading (videos, music, tv shows, etc.)
- Online shopping and online travel booking
- Online gaming and online entertainment activities

Write a report to Mr. Mark Sloan, the Senior Manager of IT Technical Specialist Sdn. Bhd. You have to elaborate on the complaints mentioned above and suggest **THREE (3)** ways to solve the problems. Elaborate on all the points given and include other relevant information. The body of the report **should not be more than 250 words.**

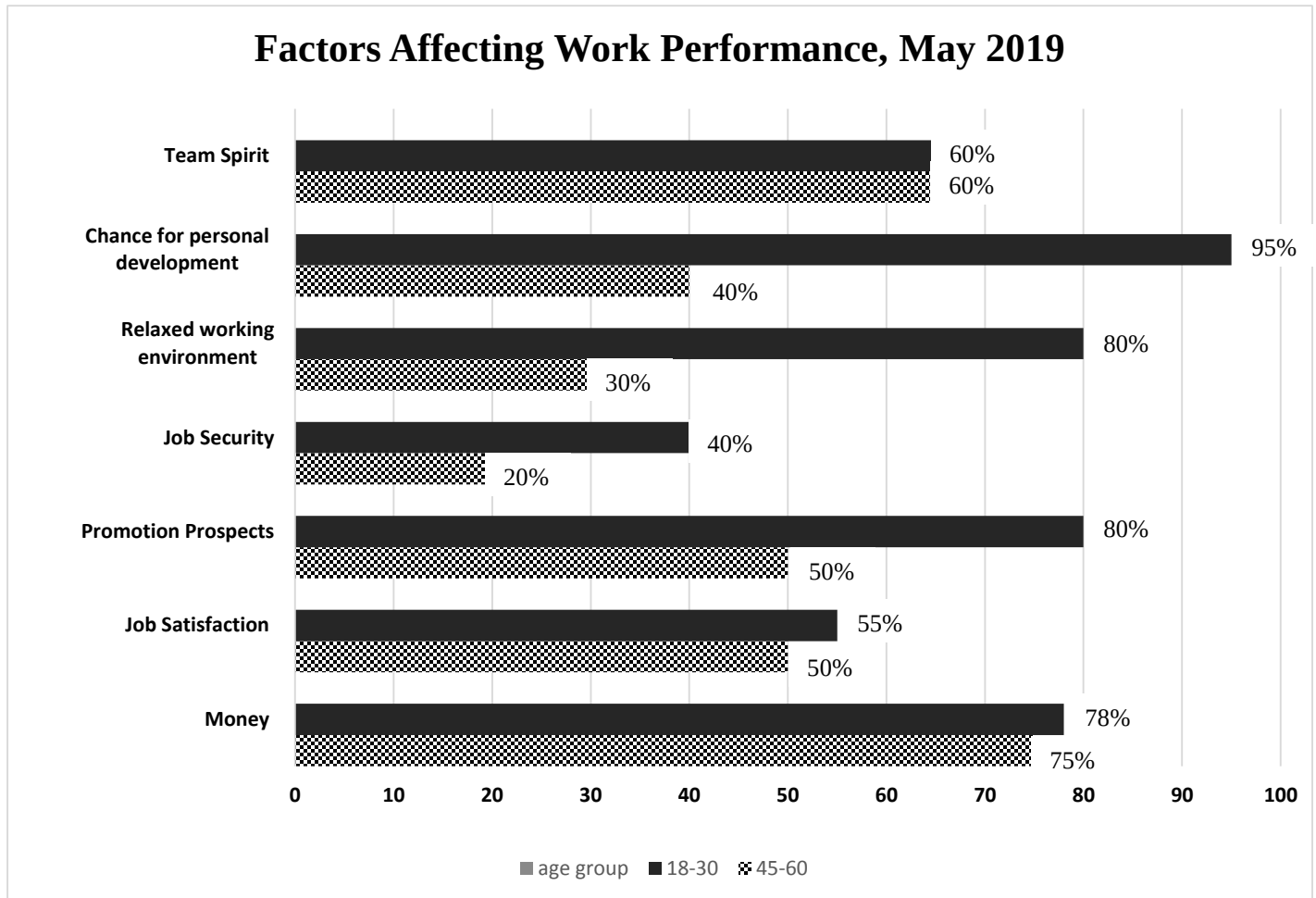
(30 marks)

[TOTAL: 30 MARKS]

MPU32143 ENGLISH FOR INFORMATION TECHNOLOGY**SECTION C [TOTAL: 30 MARKS]****INTERPRETATION OF GRAPHICAL INFORMATION**

Answer **THE** question.

Q1. Study the given bar graph. In **250** words, write about **the percentage of factors affecting work performance at ISUS company by age group, May 2019**. Interpret, describe and make comparisons between the components in the bar graph. You are **required to provide** logical and sound analysis.



(30 marks)

[TOTAL: 30 MARKS]